UNIVERSITEIT VAN PRETORIA | *UNIVERSITY OF PRETORIA*

Departement Wiskunde en Toegepaste Wiskunde | *Department of Mathematics and Applied Maths*

4 Junie / *June* 2003
Punte / *Marks*:50
Tyd / *Time*: 150 min

Punte : Intern *Marks : Internal*	Veranderinge *Changes*	Punte : Ekstern *Marks :External*

WTW 158 : CALCULUS

EKSAMEN / *EXAMINATION*

Eksterne eksaminator / *External examiner*: Prof NFJ van Rensburg

Van / *Surname* :

Voorname / *First names*:

Studentenommer / *Student number*: | | | | | | | – |

Handtekening / *Signature*:

Studierigting / *Field of Study*:	B	C	E	L	M	N	P	R	S	Z	Ander / Other

Telefoonnommer gedurende eksamenperiode / *Telephone number during exam period*	

Interne eksaminatore / Internal examiners	Interne eksaminatore / Internal examiners
Mev / Mrs A E du Preez	Dr S M Maepa
Prof JC Engelbrecht	**Prof M J Schoeman**

Lees eers die volgende instruksies

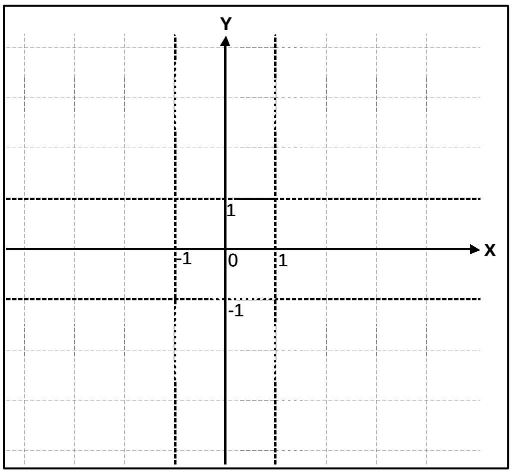
1. *Slegs nie-programmeerbare sakrekenaars mag gebruik word.*
2. *Die vraestel bestaan uit bladsye 1 tot 9 (vrae 1-17) Kontroleer of u vraestel volledig is.*
3. *Doen alle krapwerk op die blanko bladsy(e). Dit word nie nagesien nie.*
4. *As u meer as die beskikbare ruimte vir 'n antwoord nodig het, gebruik dan ook die blanko bladsy(e) en dui dit duidelik aan deur 'n **raam daarom te trek.***
5. *Geen potloodwerk word nagesien nie.*
6. *Korreksievloeistof (bv.Tipp-Ex) mag nie gebruik word nie.*
7. *Geen antwoordboek mag uit die lokaal geneem word nie*
8. ***Toon alle bewerkings duidelik***

First read the following instructions

1. *Only non-programmable calculators may be used.*
2. *The question paper consists of pages 1 to 9 (questions 1-17). Check if your paper is complete.*
3. *Scribbling can be done on the empty page(s). This will not be marked.*
4. *If you need more than the available space for an answer, you may also use the empty page(s). Indicate it clearly by **framing** it.*
5. *Work in pencil will not be marked.*
6. *Use of correction fluid (e.g.Tipp-Ex) is not allowed.*
7. *No test scripts may be removed from the venues*
9. ***Show all computations clearly***

Vraag 1 / *Question* 1

Skets die grafiek van $f(x) = \text{bgtan}(x+1)$.
Sketch the graph of $f(x) = \arctan(x+1)$.



[2]

Vraag 2 / *Question* 2

Bepaal $\frac{dy}{dx}$ as / *Determine* $\frac{dy}{dx}$ *if*

(i) $y = x\sqrt{1-x^2}$

[1]

(ii) $y = 3^x$

[1]

(iii) $x^3 - y^3 = xy$

[2]

Vraag 3 / *Question* 3

Bepaal / *Evaluate*

(i) $\int \frac{x^2}{x^3-4}dx$

[1]

(ii) $\int \left(\frac{1}{x^2} - \cos x \right) dx$

[1]

(iii) $\int \cosh (2x+3)\, dx$

[1]

(iv) $\int \frac{\sin\sqrt{x}}{\sqrt{x}}dx$

[1]

Vraag 4 / *Question* 4

Bepaal die vergelyking van die raaklyn aan die grafiek van $f(x) = \text{bgsin}x$ by $x = \frac{1}{2}$.

Find the equation of the tangent line to the graph of $f(x) = \arcsin x$ *at* $x = \frac{1}{2}$.

[3]

Vraag 5 / *Question* 5

Bepaal 'n formule vir $f^{(n)}(x)$ as $f(x) = \ln(x-1)$, $n \in \mathbb{N}$.

Find a formula for $f^{(n)}(x)$ *if* $f(x) = \ln(x-1)$, $n \in \mathbb{N}$.

[2]

Vraag 6 / *Question* 6

Laat / *Let* $f(x) = \dfrac{x}{(x-1)^2}$, $\quad f'(x) = \dfrac{-(x+1)}{(x-1)^3}$ $\quad$ en / *and* $f''(x) = \dfrac{2(x+2)}{(x-1)^4}$.

Toon alle berekeninge en bepaal die
Show all computations and determine the

(*i*) vertikale asimptoot van *f*.
vertical asymptote of f.

[1]

(ii) horisontale asimptoot van f.
horizontal asymptote of f.

[2]

(iii) lokale minimumwaarde van f.
local minimum value of f.

[2]

(iv) interval(le) waarop f konkaaf na bo is.
interval(s) where f *is concave upward*.

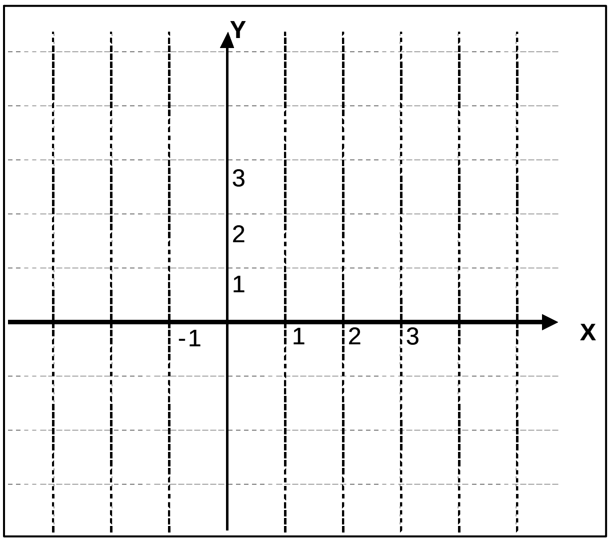
[2]

Vraag 7 / *Question* 7

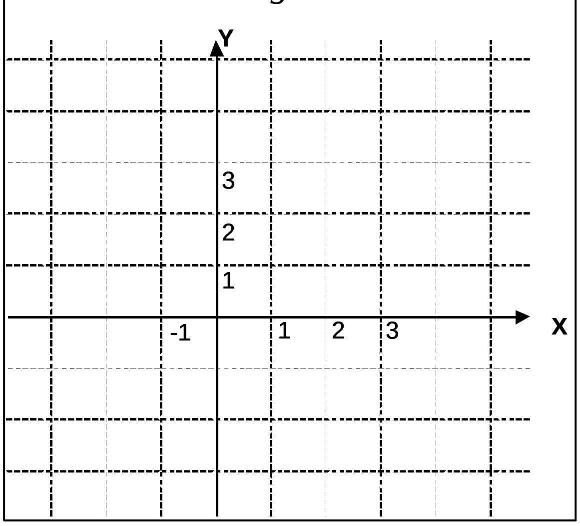
Skets 'n funksie f met die volgende eienskappe :

Sketch a function f with the following properties:

- $\boxed{f(0) = 1, \quad f'(0) = 0}$
- $\boxed{f'(x) < 0 \text{ as / } \textit{if } 0 < x < 3}$
- $\boxed{f'(x) > 0 \text{ as / } \textit{if } x < 0 \text{ en / } \textit{and } x > 3}$
- $\boxed{f''(x) < 0 \text{ as / } \textit{if } x \neq 3}$
- $\boxed{\lim_{x \to 3} f(x) = -\infty}$



Rofwerk / *Rough work*



[3]

Vraag 8 / *Question* 8

Gegee / *Given* $\quad f(x) = \begin{cases} \dfrac{x^2 - a^2}{x - a}, & \text{as / } \textit{if } x \neq a \\ 4, & \text{as / } \textit{if } x = a \end{cases}$

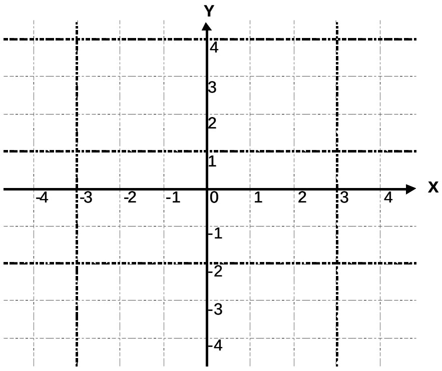
(i) Bepaal die waarde van a waarvoor f kontinu is op $\mathbb{R}$.

Determine the value of a such that f is continuous on $\mathbb{R}$.

[2]

(ii) Skets die grafiek van f vir hierdie a.

Sketch the graph of f for this a.



[1]

Vraag 9 / *Question* 9

Bepaal / *Evaluate* $\int \sin^2 x \cos^3 x dx$

[3]

Vraag 10 / *Question* 10

Die grafiek van $y = f(x)$ gaan deur $(1,-1)$. As $f'(3x) = x$, bepaal $f(x)$.

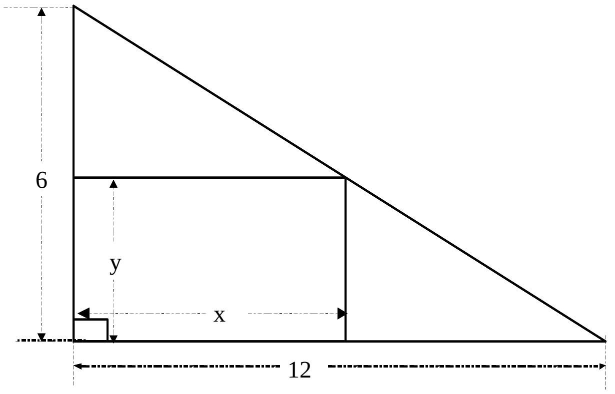
The graph of $y = f(x)$ passes through $(1,-1)$. If $f'(3x) = x$, determine $f(x)$.

[3]

Vraag 11 / *Question* 11

Bepaal die afmetings van die reghoek met maksimum oppervlakte wat in 'n reghoekige driehoek met afmetings soos in die gegewe skets, ingeskryf kan word.

Determine the dimensions of the rectangle with maximum area that can be inscribed in a right angled triangle with dimensions as in the given sketch.



[4]

Vraag 12 / *Question* 12

Bepaal 'n eenheidsvektor loodreg (ortogonaal) op beide $\overline{a} = \overline{i} + \overline{j}$ en $\overline{b} = \overline{i} + \overline{k}$.

Determine a unit vector perpendicular (orthogonal) to both $\overline{a} = \overline{i} + \overline{j}$ *and* $\overline{b} = \overline{i} + \overline{k}$.

[2]

Vraag 13 / *Question* 13

Bepaal die vektorprojeksie van $\overline{a} = \overline{i} + \overline{k}$ op $\overline{b} = \overline{i} - \overline{j}$.

Determine the vector projection of $\overline{a} = \overline{i} + \overline{k}$ *on* $\overline{b} = \overline{i} - \overline{j}$.

[2]

Vraag 14 / *Question* 14

Bepaal simmetriese vergelykings van die lyn deur die oorsprong en wat ewewydig is aan die lyn met parametriese vergelykings $x = 1 - 2t, \quad y = t, \quad z = 2 + t, \quad t \in R.$

Determine symmetric equations for the line through the origin and parallel to the line with parametric equations $x = 1 - 2t, \quad y = t, \quad z = 2 + t, \quad t \in R.$

[2]

Vraag 15 / *Question* 15

Bepaal 'n vergelyking van die vlak deur die punt $(-1,1,2)$ en loodreg op die lyn met vergelykings $x = 2, \quad \dfrac{y-1}{3} = \dfrac{z+1}{-2}$.

Determine an equation for the plane through the point $(-1,1,2)$ *and perpendicular to the line with equations* $x = 2, \quad \dfrac{y-1}{3} = \dfrac{z+1}{-2}$.

[2]

Vraag 16 / *Question* 16

Bepaal 'n vergelyking van die vlak deur die punt $(-1,6,5)$ en ewewydig aan die vlak $2x - 3y + z = 4$.

Determine an equation for the plane through the point $(-1,6,5)$ *and parallel to the plane* $2x - 3y + z = 4$.

[2]

Vraag 17 / *Question* 17

Beskryf (of skets) die gebied in R^3 wat voorgestel word deur $x^2 + y^2 = 4$.

Describe (or sketch) the region in R^3 *that is represented by* $x^2 + y^2 = 4$.

[2]